

PANDAS DATAFRAME & BASIC STATISTICS

ERT 474/574

Open-Source Hydro Data Analytics

Sep 16th 2026



Logistics

- HW2 due next Tuesday



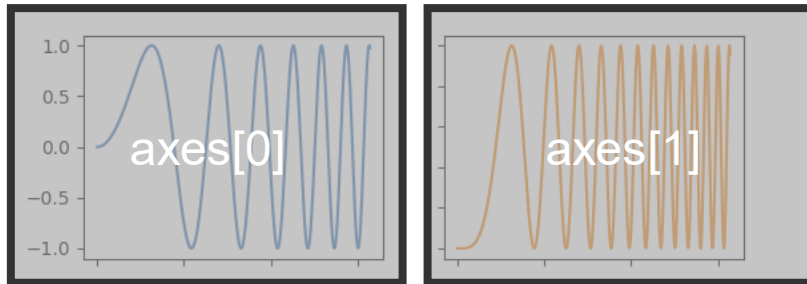
Recap - Numpy

- How do we import numpy packages?
- Given an array `x=np.array([1,2,3])`, how do we add 3 to every element in `x`?
- What is the numpy function to generate a continuous array, such as `[0,1,2,3,4,5]`?
- What is the difference between `math.exp` and `np.exp`?
- What would happen if we convert a mixed-type list to an array, `my_list = ['Yifan', 100, 'apple']`?
- `import numpy as np`
- `x+3`
- `np.arange(6)` or `np.linspace(0,5,6)`
- `math.exp` can only be used to calculate the exponential for one value while `np.exp` can be used for an array
- The array would be `['Yifan', '100', 'apple']`

Recap - matplotlib

- How to import the main interface to actual plotting functions?
- `plt.figure` versus `plt.subplots`

- What's the syntax if we want two subplots (1 row, 2 columns)?



- What is the syntax to do a line plot?
- What is the syntax to do a scatter plot?

- `import matplotlib.pyplot as plt`
- `plt.figure` only creates the figure and you manually add subplots, which is usually used to make single figure. `plt.subplots` creates both the figure and subplots in one call, making it simpler for layouts with multiple plots.

- `fig, axes=plt.subplots(1,2)`

- `plt.plot()`

- `plt.scatter()`

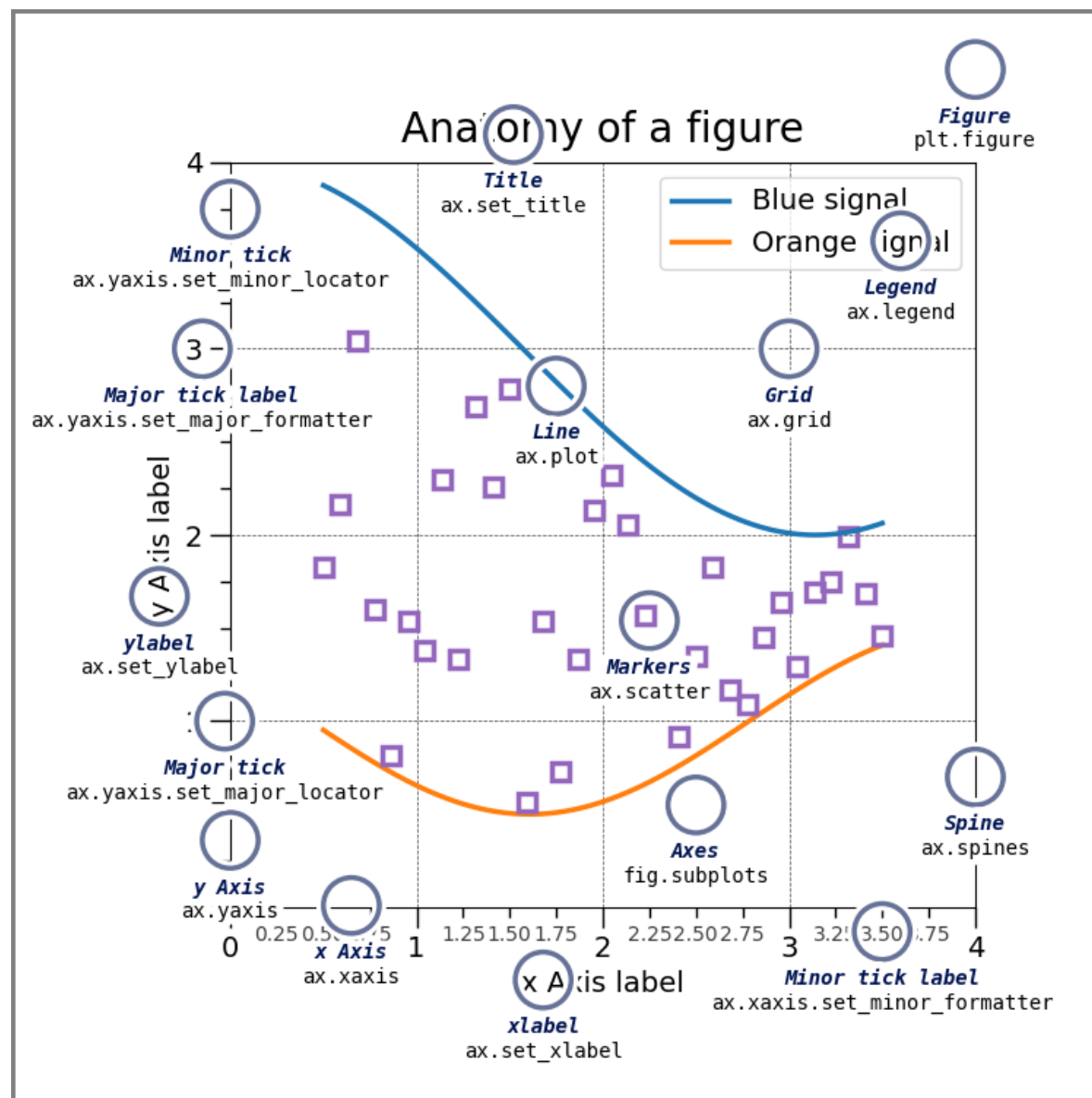
Recap - matplotlib

- Anatomy of a figure!

- Credit:

<https://matplotlib.org/stable/gallery/showcase/anatomy.html>

```
fig, ax=plt.subplots(1,1)
```



Last week, we introduced **Numpy**
(a powerful tool to generate data
arrays and calculations)

How can we more effectively
manipulate data?

```
import pandas as pd
```

Data Structure

Powerful functions

Data Structure

Data Structure	Dimension
Series	1
Data Frames	2

What is the difference between series and Data Frames?

Data Series

tom	105
bob	306
nancy	3560
dan	1200
eric	50

Data Framework

	Fav_number	Fav_color
tom	105	red
bob	306	blue
nancy	3560	orange
dan	1200	pink
eric	50	green

DataFrame syntax

	column	
	Fav_number	Fav_color
Row	tom	105 red
	bob	306 blue
	nancy	3560 orange
	dan	1200 pink
	eric	50 green

DataFrame syntax

	Column name	
	Fav_number	Fav_color
index	tom	105 red
	bob	306 blue
	nancy	3560 orange
	dan	1200 pink
	eric	50 green

DataFrame syntax

`df.loc[index, column name]`

Column name

index

	Fav_number	Fav_color
tom	105	red
bob	306	blue
nancy	3560	orange
dan	1200	pink
eric	50	green

DataFrame syntax

`df.loc['tom','Fav_number']`

Column name

index

	Fav_number	Fav_color
tom	105	red
bob	306	blue
nancy	3560	orange
dan	1200	pink
eric	50	green

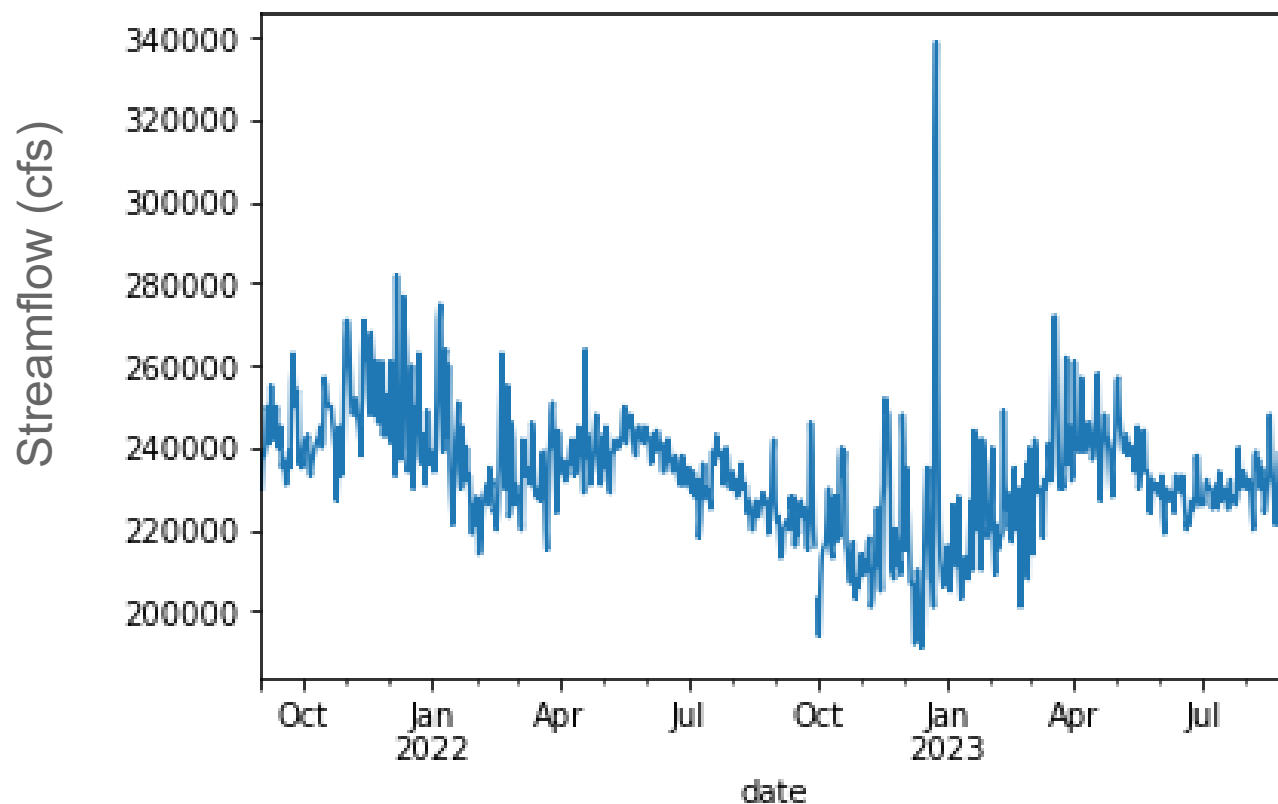
DataFrame syntax

`df.iloc[0,0]`

	0 th column	1 st column
	Fav_number	Fav_color
0 th row	tom	105
1 st row	bob	306
2 nd row	nancy	3560
3 rd row	dan	1200
4 th row	eric	50

Data manipulation for time series data

Streamflow for Niagara River @ Buffalo, NY



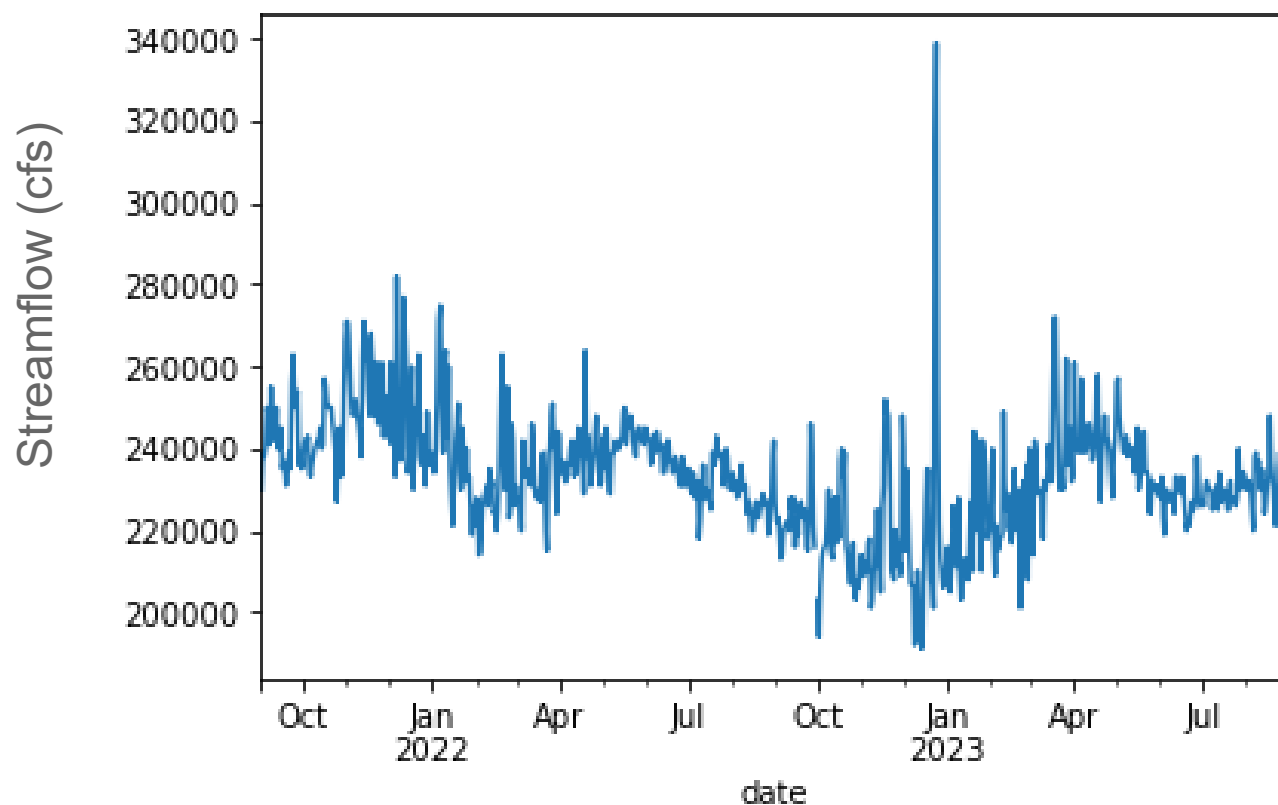
Out[70]:

	streamflow	quality_flag
date		
2022-10-01	203000.0	A
2022-10-02	194000.0	A
2022-10-03	206000.0	A
2022-10-04	213000.0	A
2022-10-05	215000.0	A
...	...	...
2022-12-28	209000.0	A
2022-12-29	206000.0	A
2022-12-30	207000.0	A
2022-12-31	211000.0	A
2023-01-01	216000.0	A

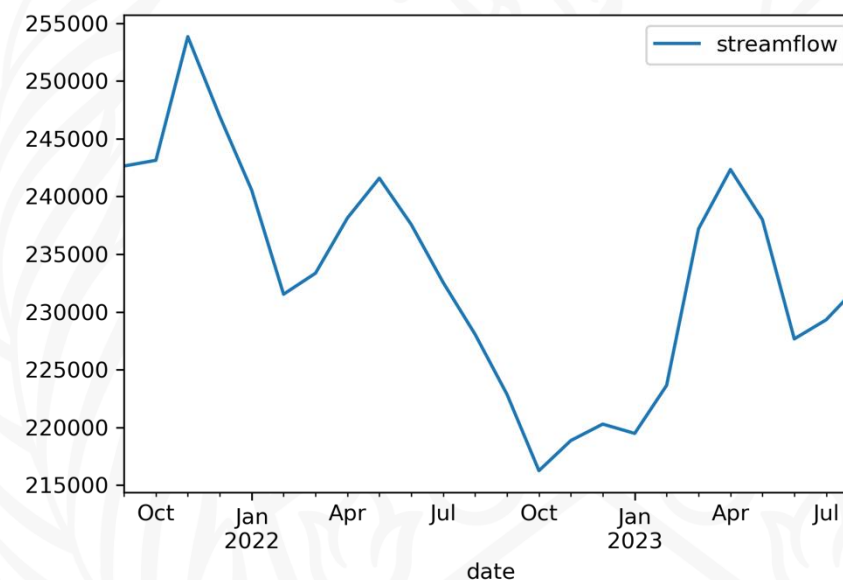
93 rows x 2 columns

Data manipulation for time series data

Streamflow for Niagara River @ Buffalo, NY

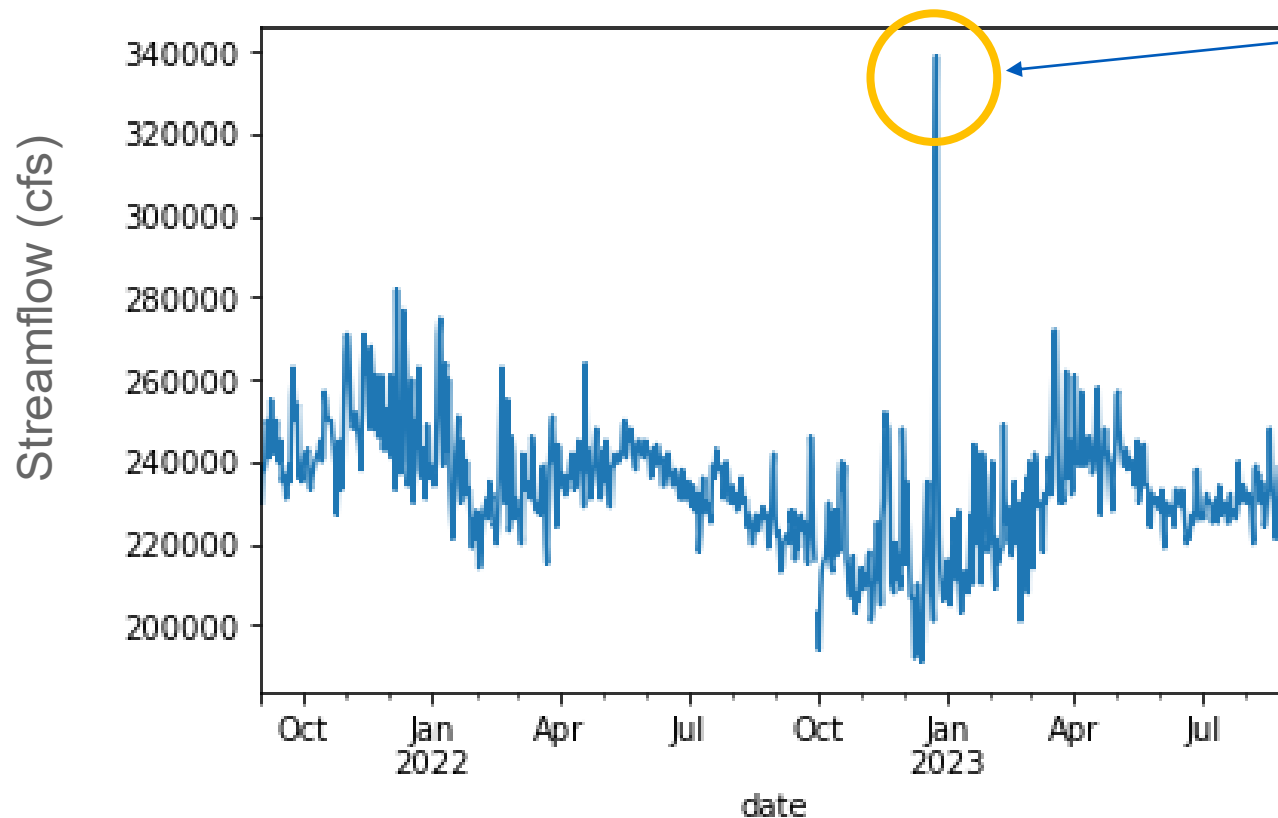


How can we change the frequency of data from daily to monthly?



Data manipulation for time series data

Streamflow for Niagara River @ Buffalo, NY



How can we identify the extreme high flow events?

When is that event?

We will practice more in next week's lab session!

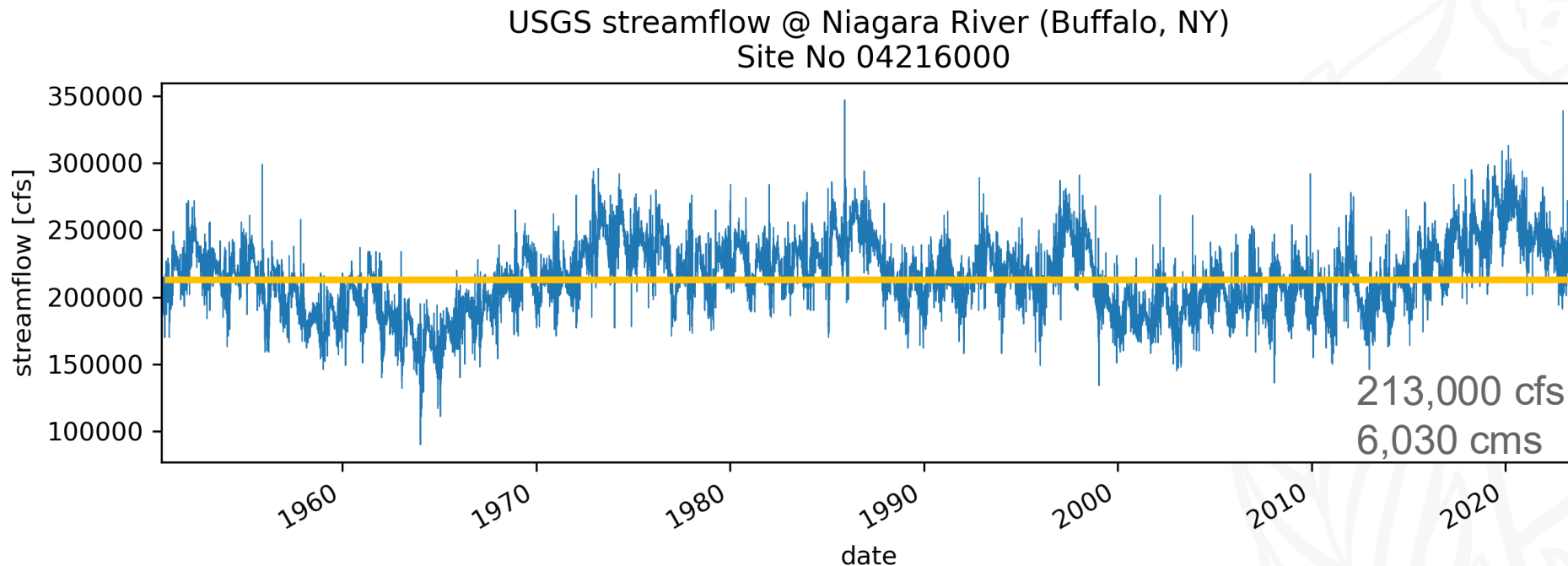
Statistical methods are widely used in hydrologic modeling

How do we do descriptive analysis when we get a data?

- Mean, variance, standard deviation (Box plot)
- PDF(Histogram), CDF (Quantile mapping), median (inter-quantile range)
- Extreme detection
 - Z-score
 - 7Q10



If you were a state hydrologist, when you were asked to give a high-level introduction to Niagara Rivers at Buffalo, what information would you provide based on the streamflow observations?

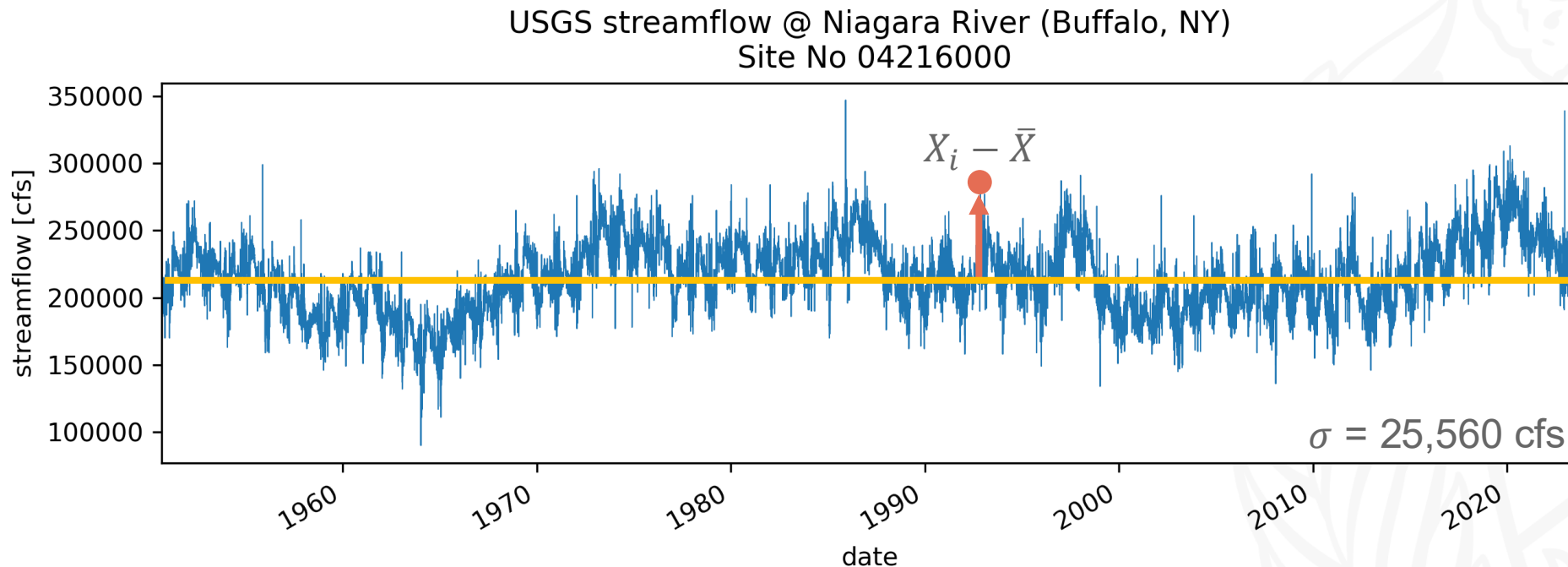


1. Sample Mean

$$\bar{X} = \sum_{i=1}^n \frac{X_i}{n}$$

Mean streamflow usually is used to evaluate the overall water availability for a region.

If you were a state hydrologist, when you were asked to give a high-level introduction to Niagara Rivers at Buffalo, what information would you provide based on the streamflow observations?



2. Sample Variance

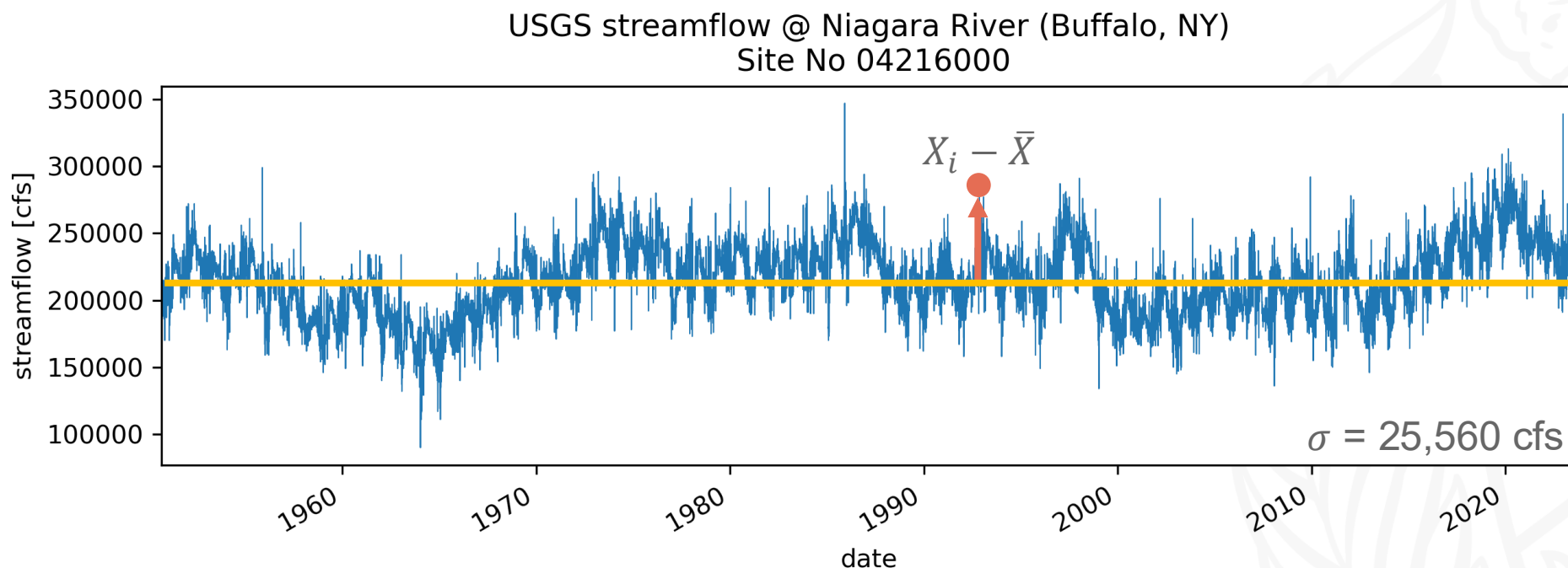
$$\sigma^2 = \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{n - 1}$$

3. Standard Deviation

$$\sigma = \sqrt{\sigma^2}$$

"Variance" refers to a measure of how spread out a set of data is from its mean (average), essentially indicating how much variation exists within a data set

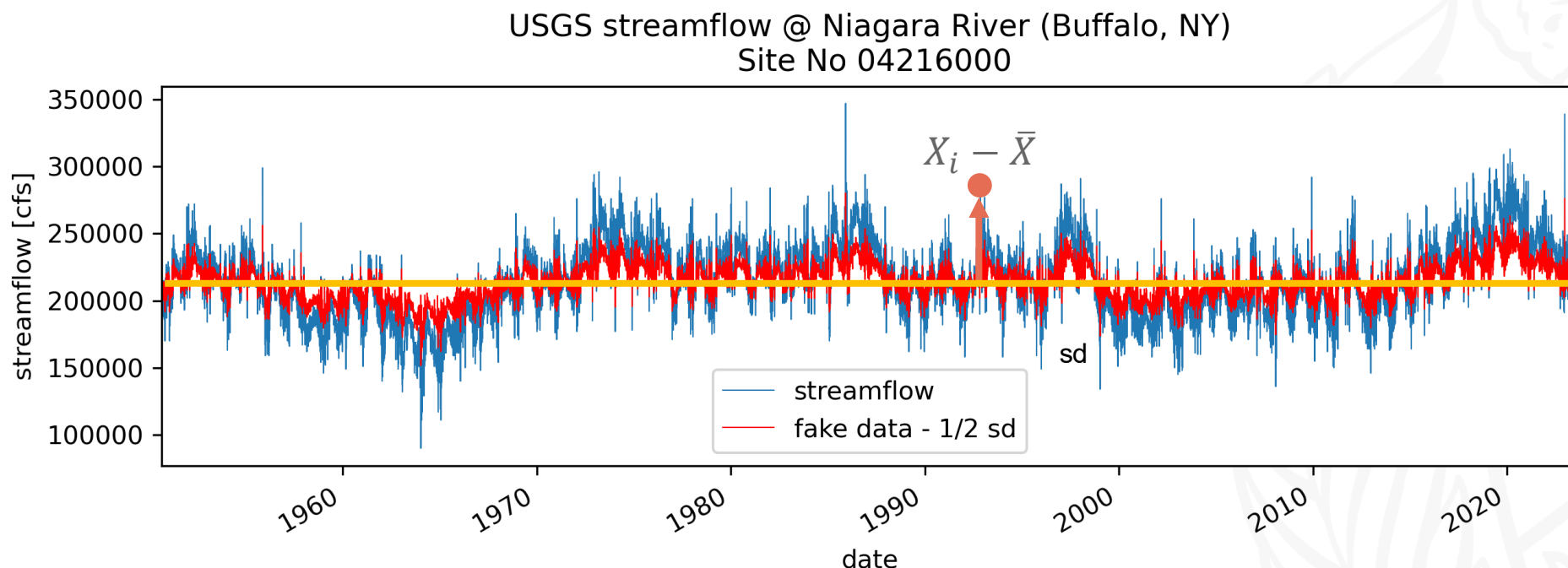
If you were a state hydrologist, when you were asked to give a high-level introduction to Niagara Rivers at Buffalo, what information would you provide based on the streamflow observations?



How do we create a time series with same $\bar{X}$ but 0.5σ ?

$$X_{i,fake} = \bar{X} + \frac{1}{2} (X_i - \bar{X})$$

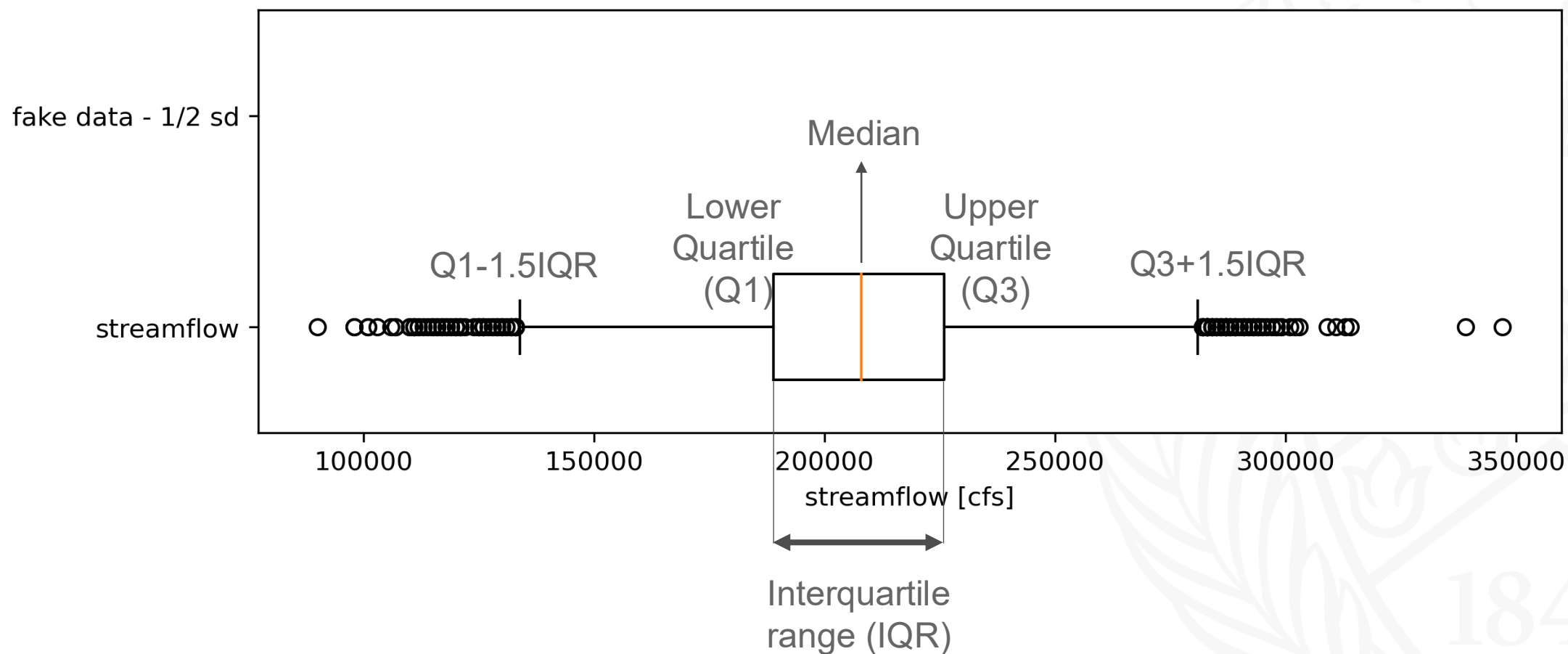
If you were a state hydrologist, when you were asked to give a high-level introduction to Niagara Rivers at Buffalo, what information would you provide based on the streamflow observations?



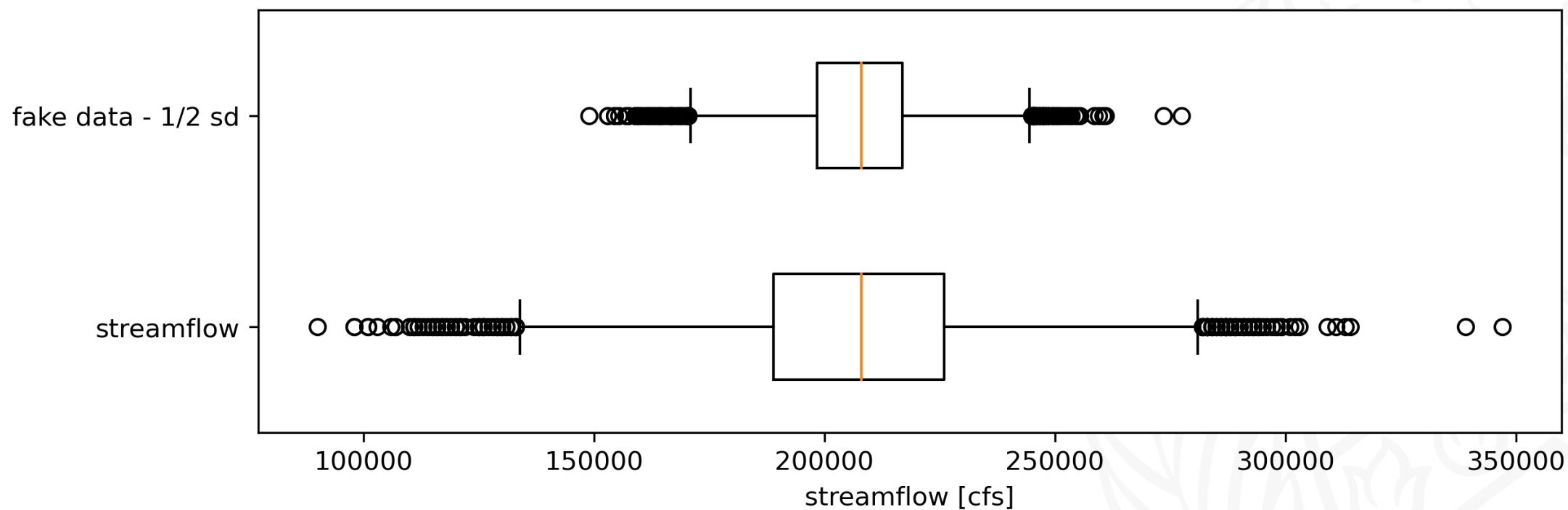
A more straightforward way to visualize the spread of the dataset?

Flow with lower standard deviation are more centered around the mean value!

Box-plot



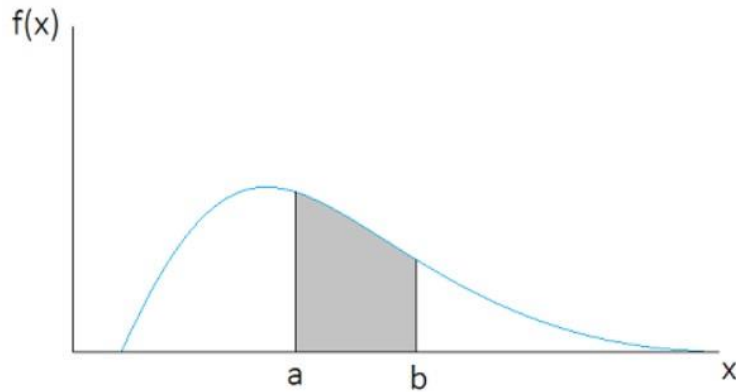
Box-plot



How do we evaluate the distribution of streamflow data?

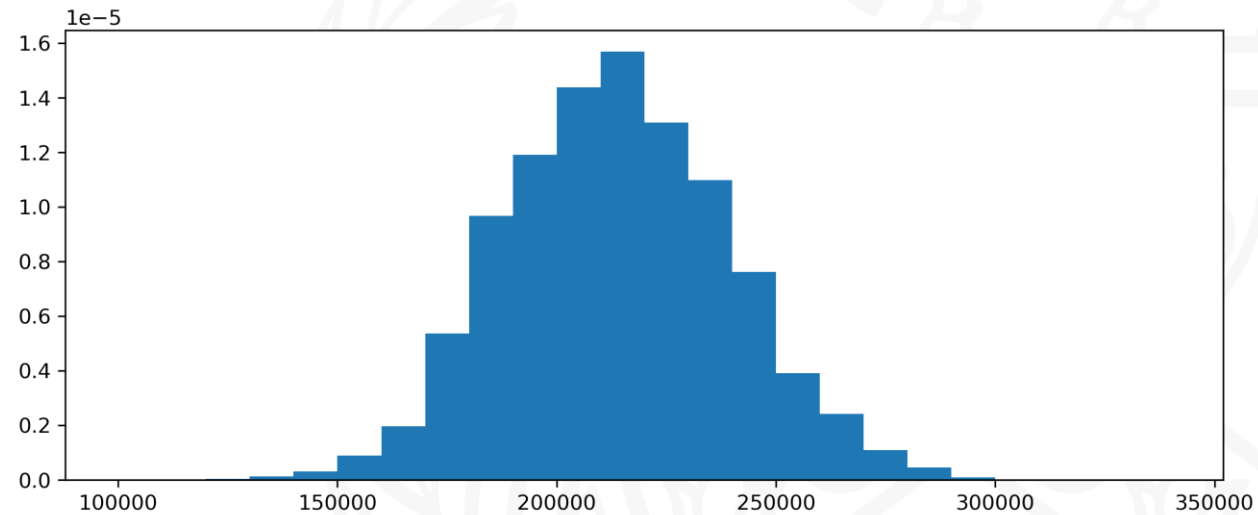
- Probability Density Function

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$



The total area below a PDF is 1.

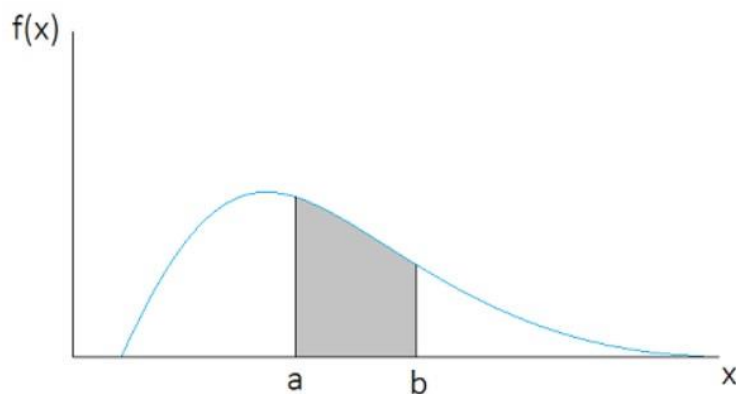
We can use histogram to visualize the PDF for time series data.



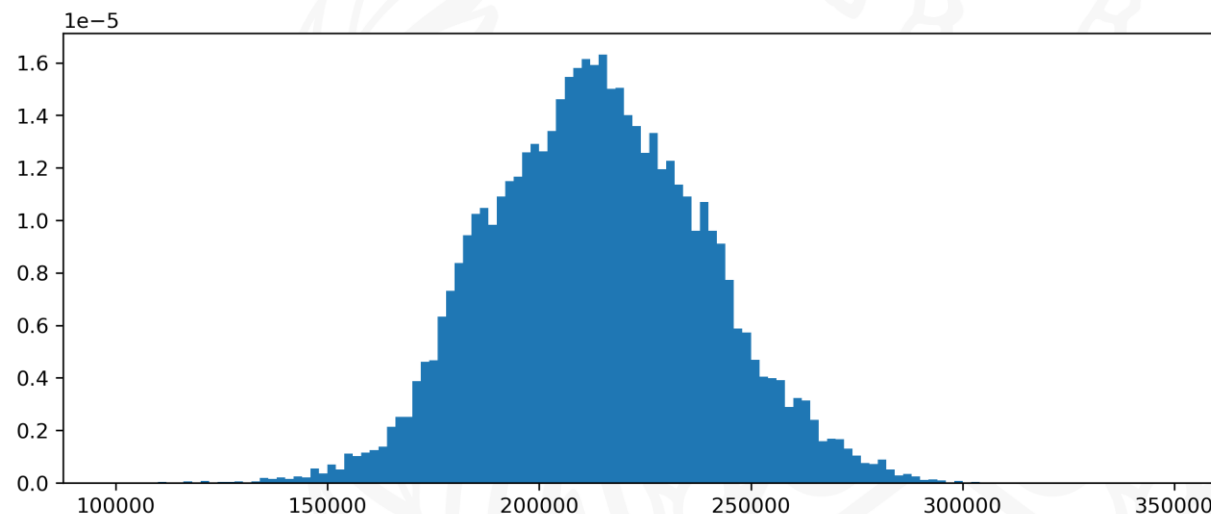
How do we evaluate the distribution of streamflow data?

- Probability Density Function

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$



We can use histogram to visualize the PDF for time series data



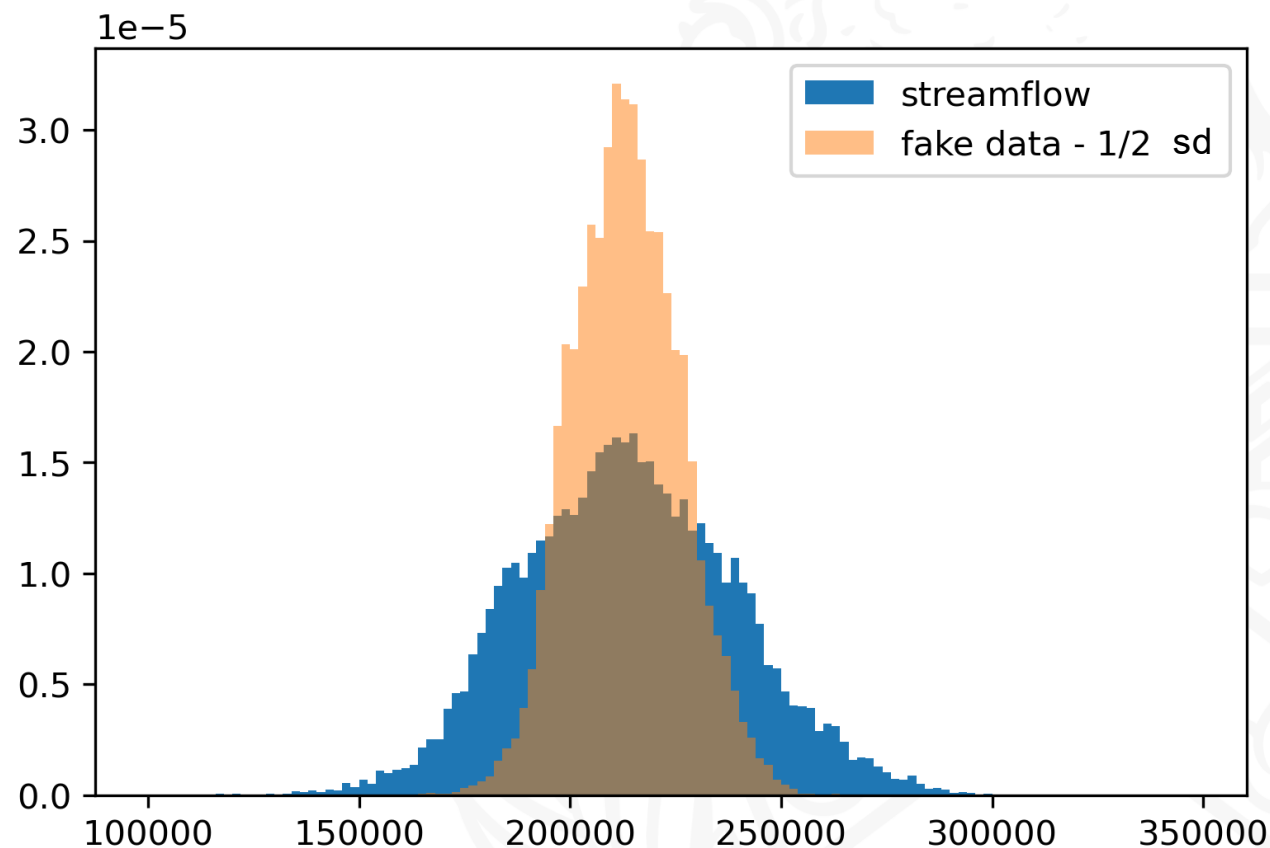
The histogram might look a bit different after we change the bin size

How do we evaluate the distribution of streamflow data?

- Probability Density Function

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

What does the PDF for our fake streamflow data look like?

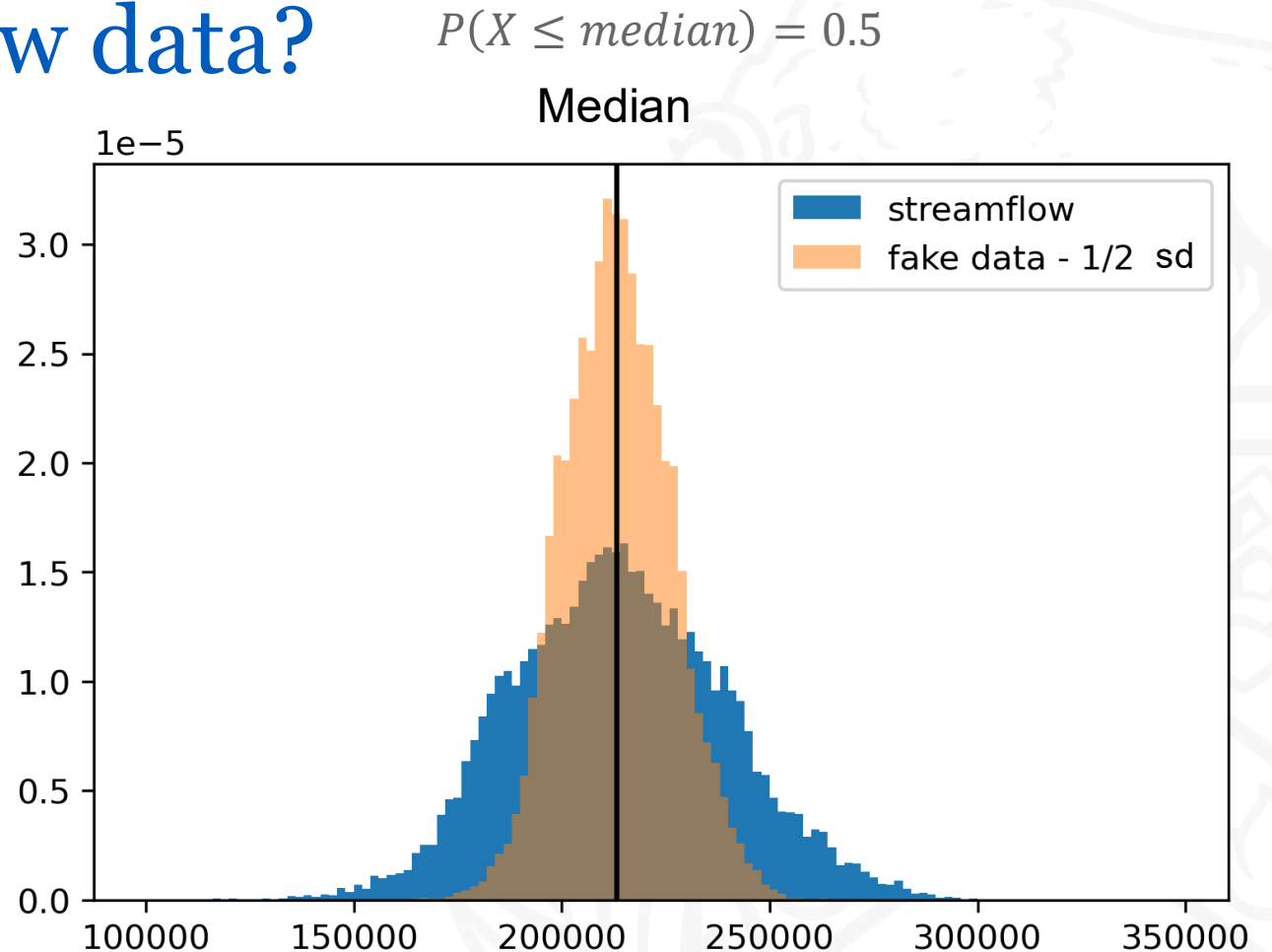


How do we evaluate the distribution of streamflow data?

- Probability Density Function

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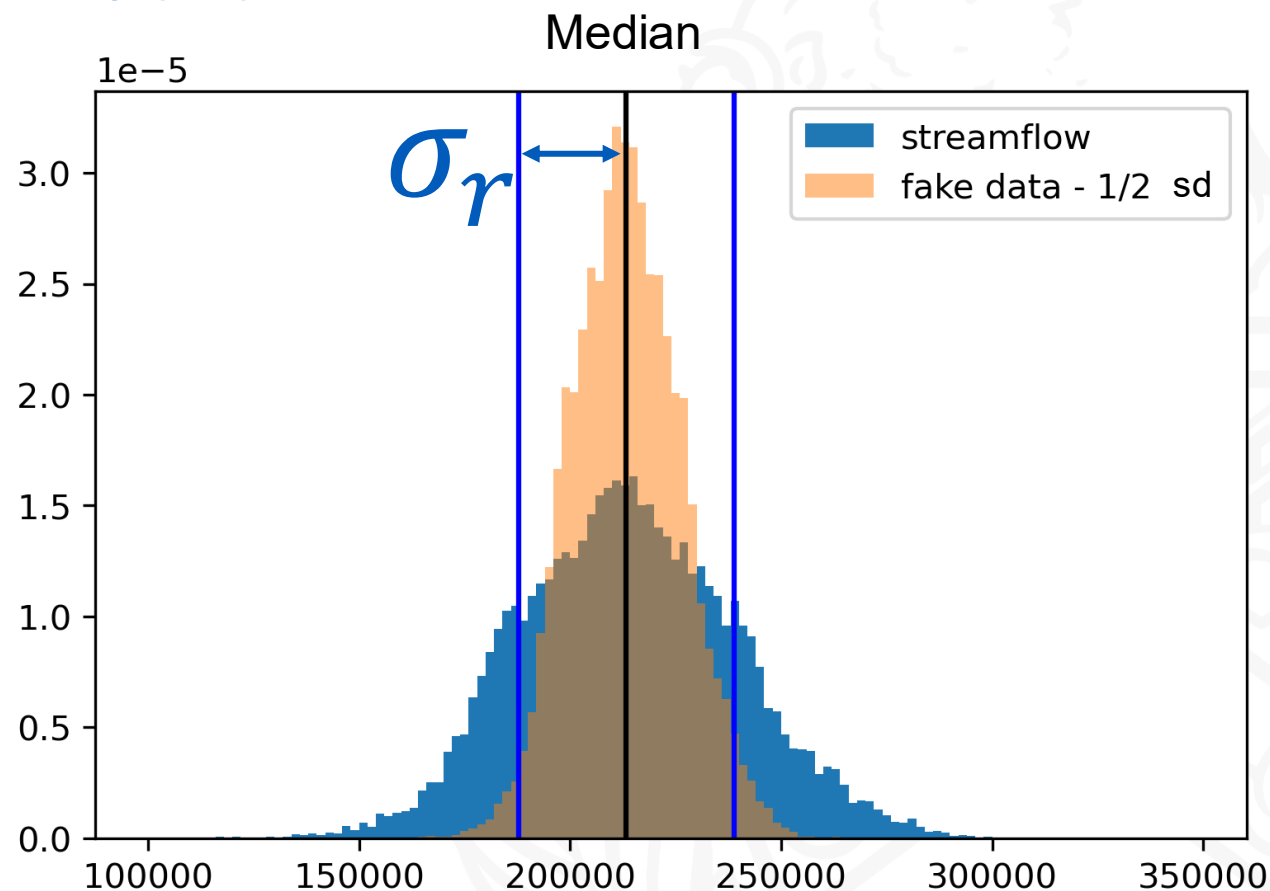


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- Probability Density Function

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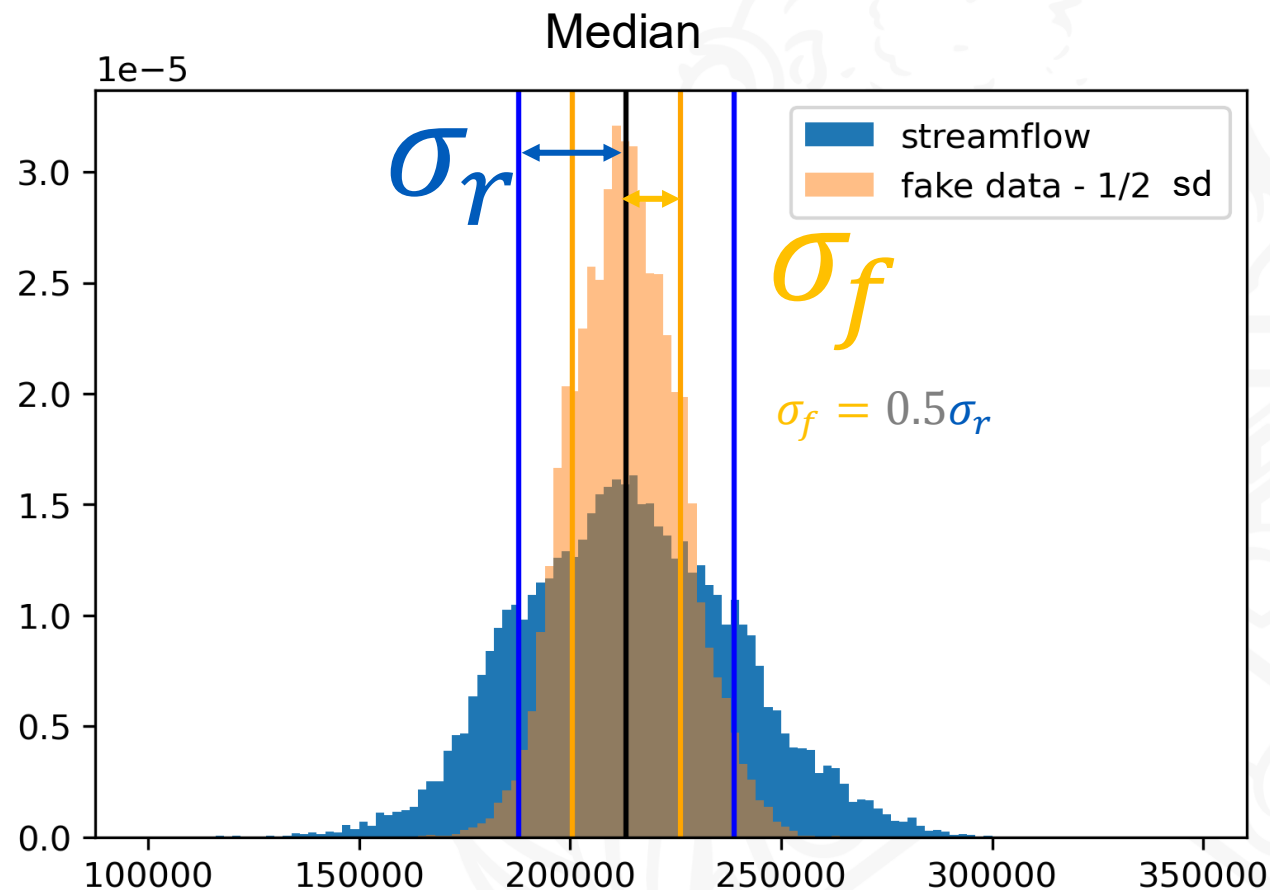


How do we evaluate the distribution of streamflow data?

- Probability Density Function

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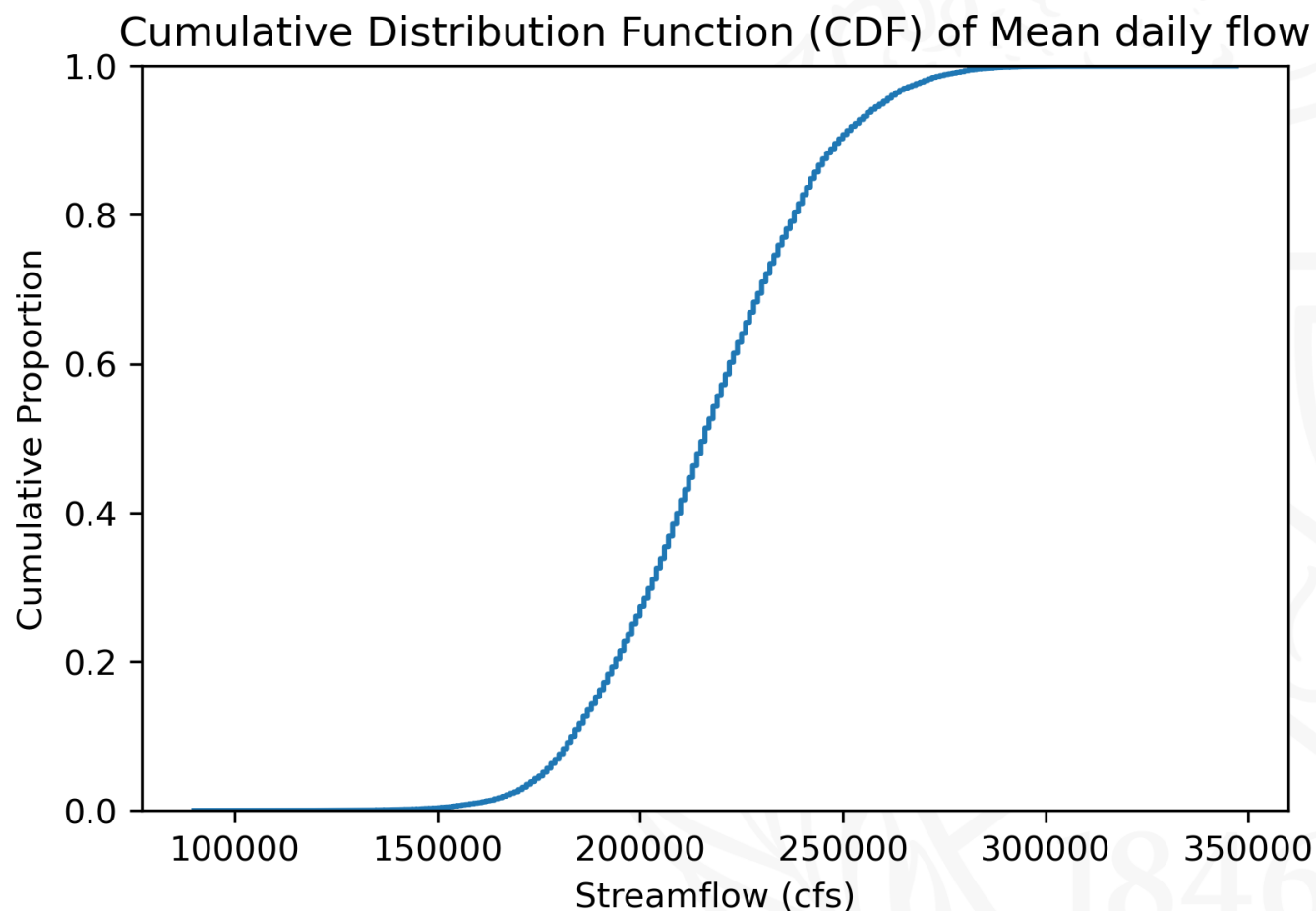
What does the PDF for our fake streamflow data look like?



How do we evaluate the distribution of streamflow data?

- Cumulative Distribution Functions

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$$

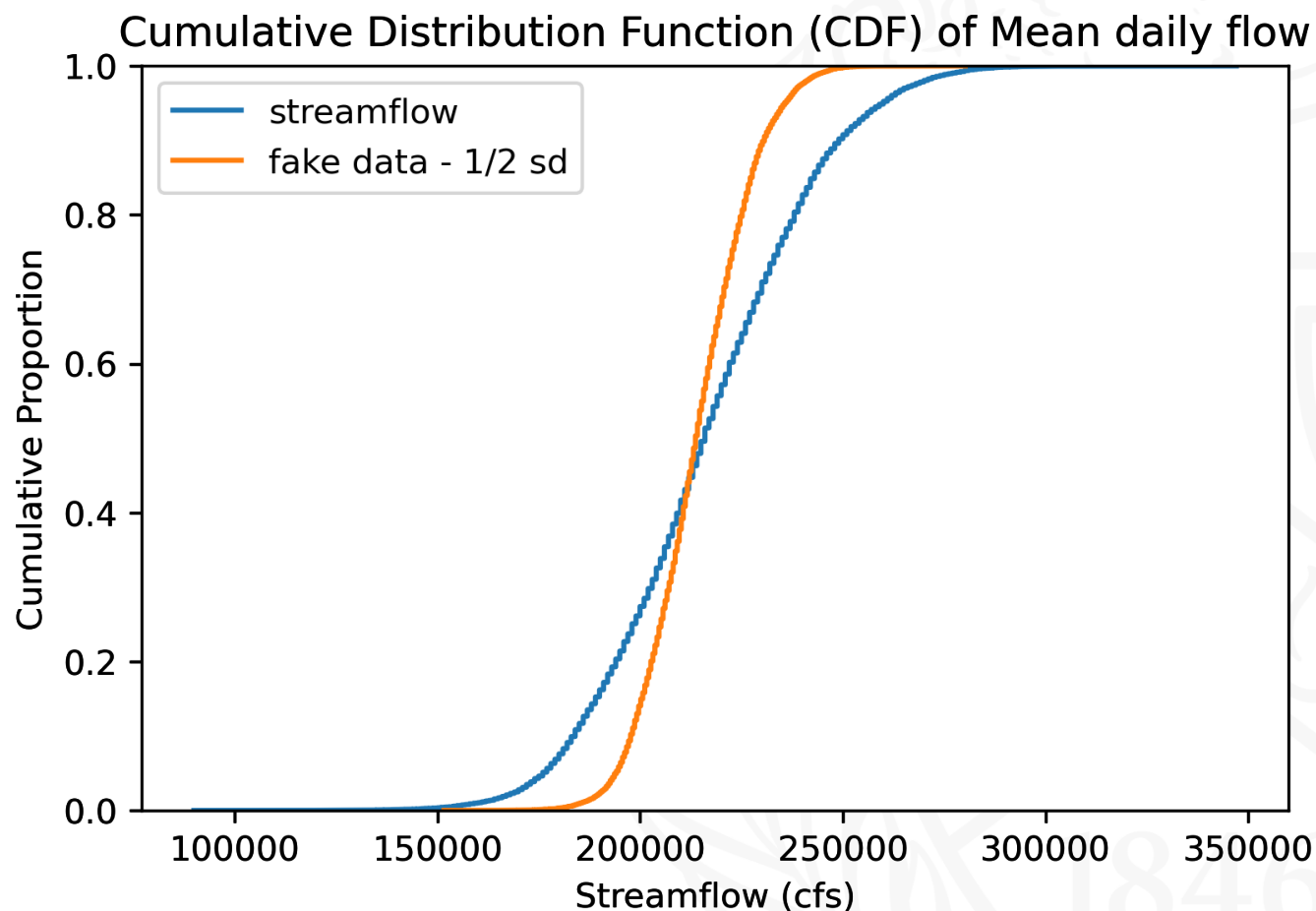


How do we evaluate the distribution of streamflow data?

- Cumulative Distribution Functions

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$$

What does the CDF for our fake streamflow data look like?

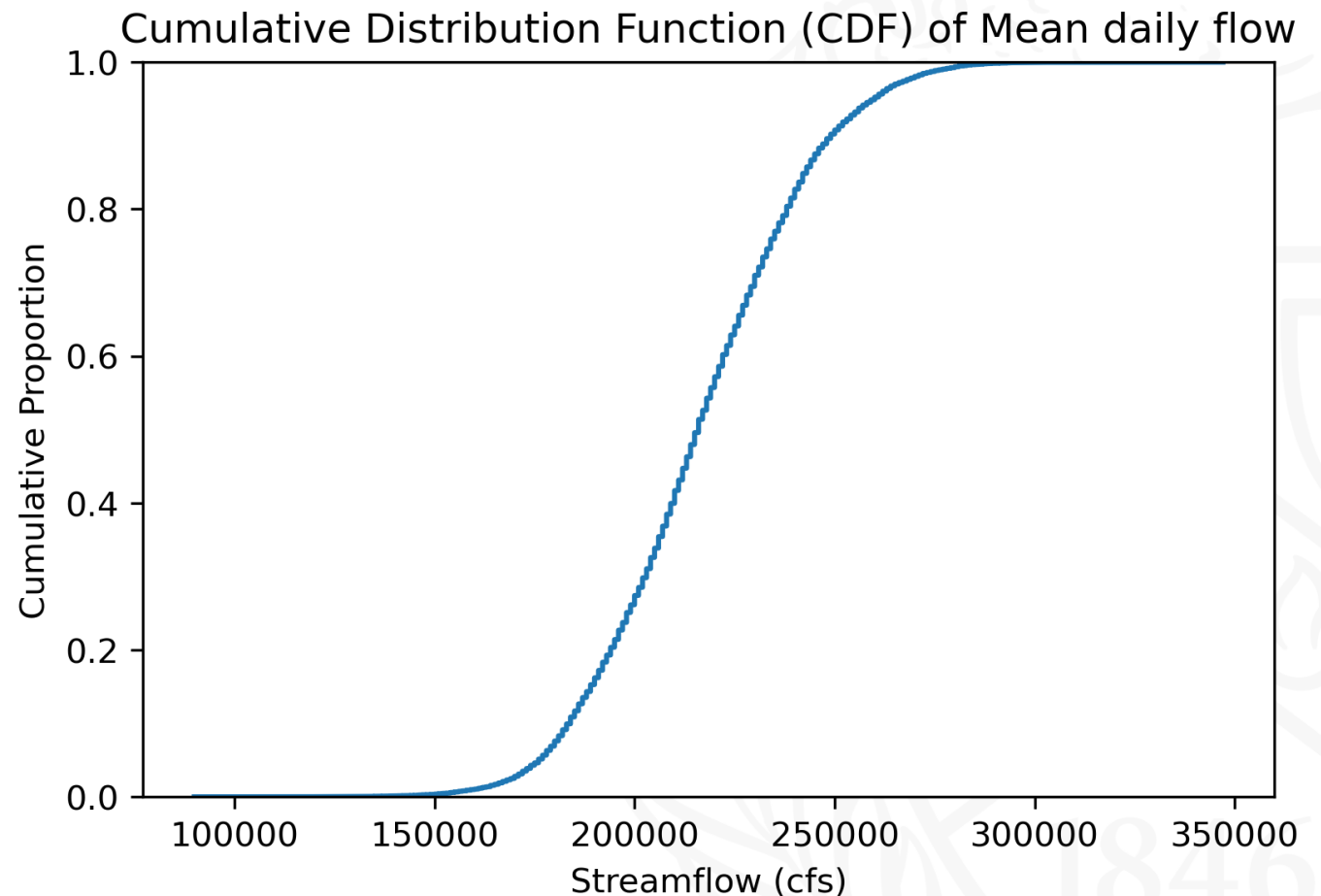


How do we evaluate the distribution of streamflow data?

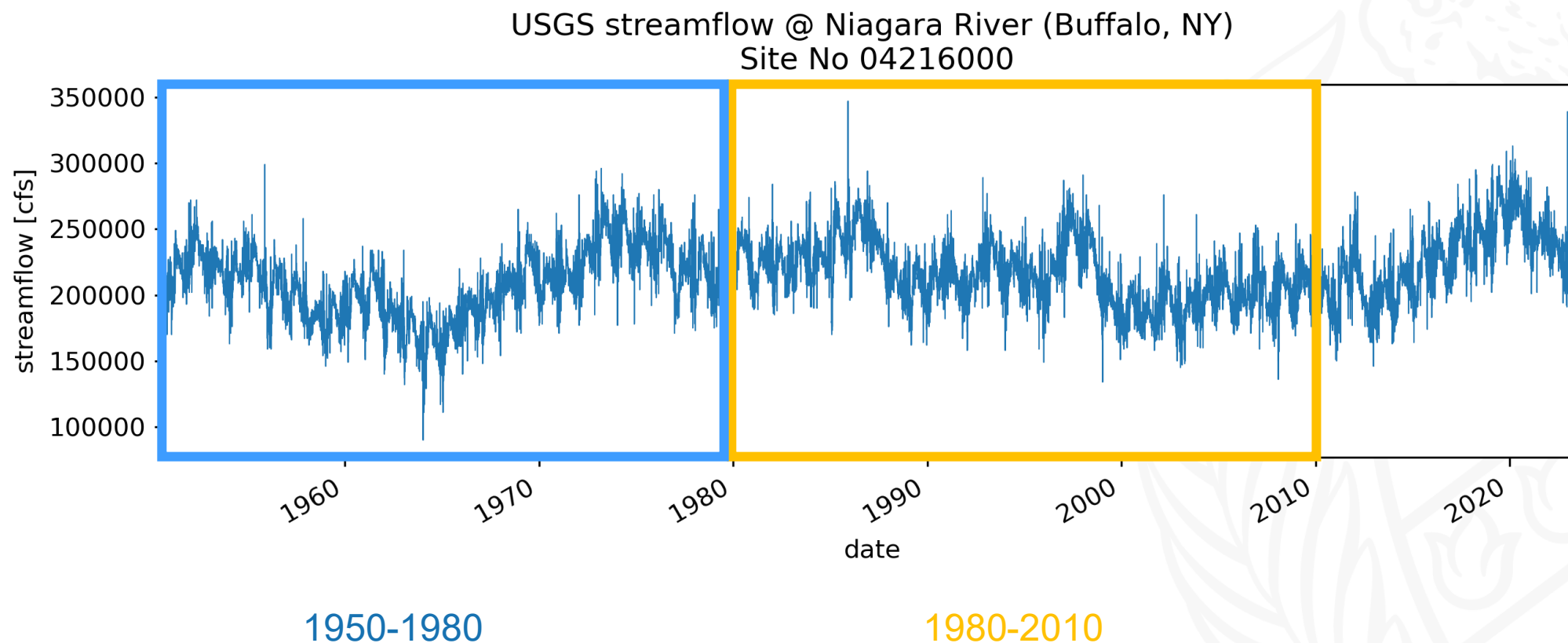
- Cumulative Distribution Functions

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) dy$$

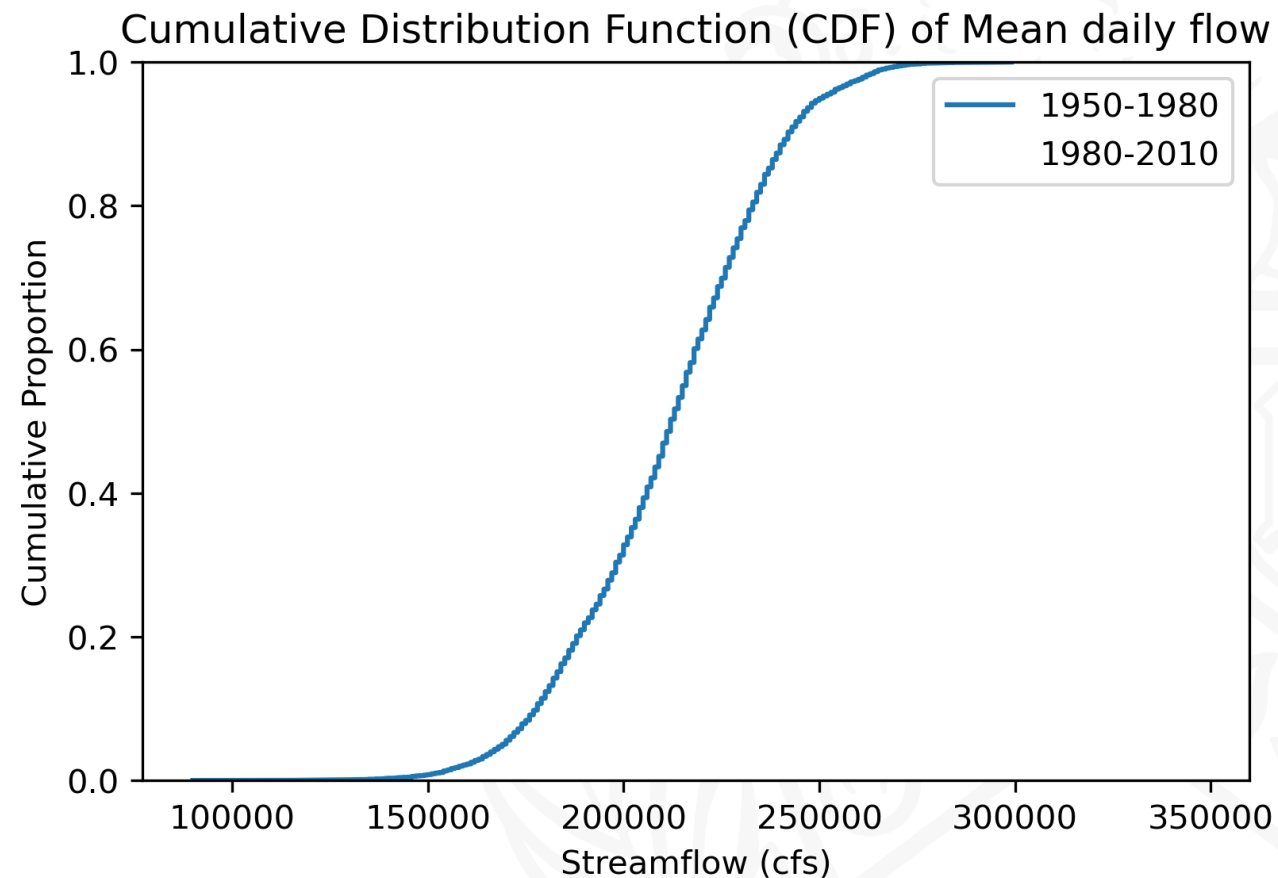
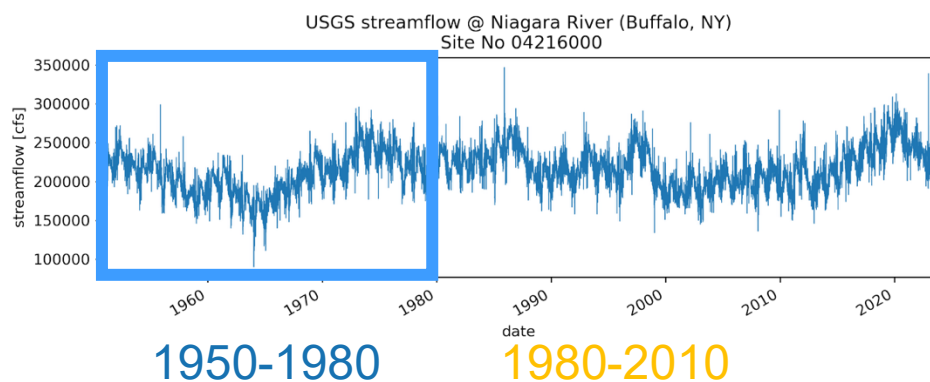
- In the analysis, CDF can be a powerful tool for visualizing the shifting of hydrologic regimes.



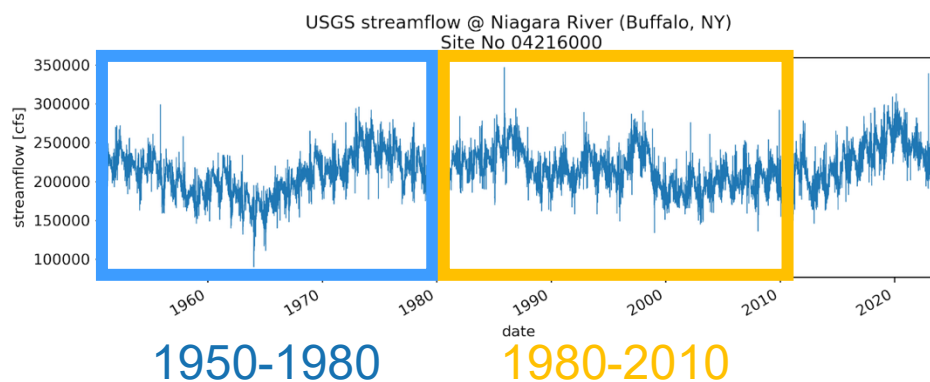
Use CDF to visualize the shift in hydrologic regimes



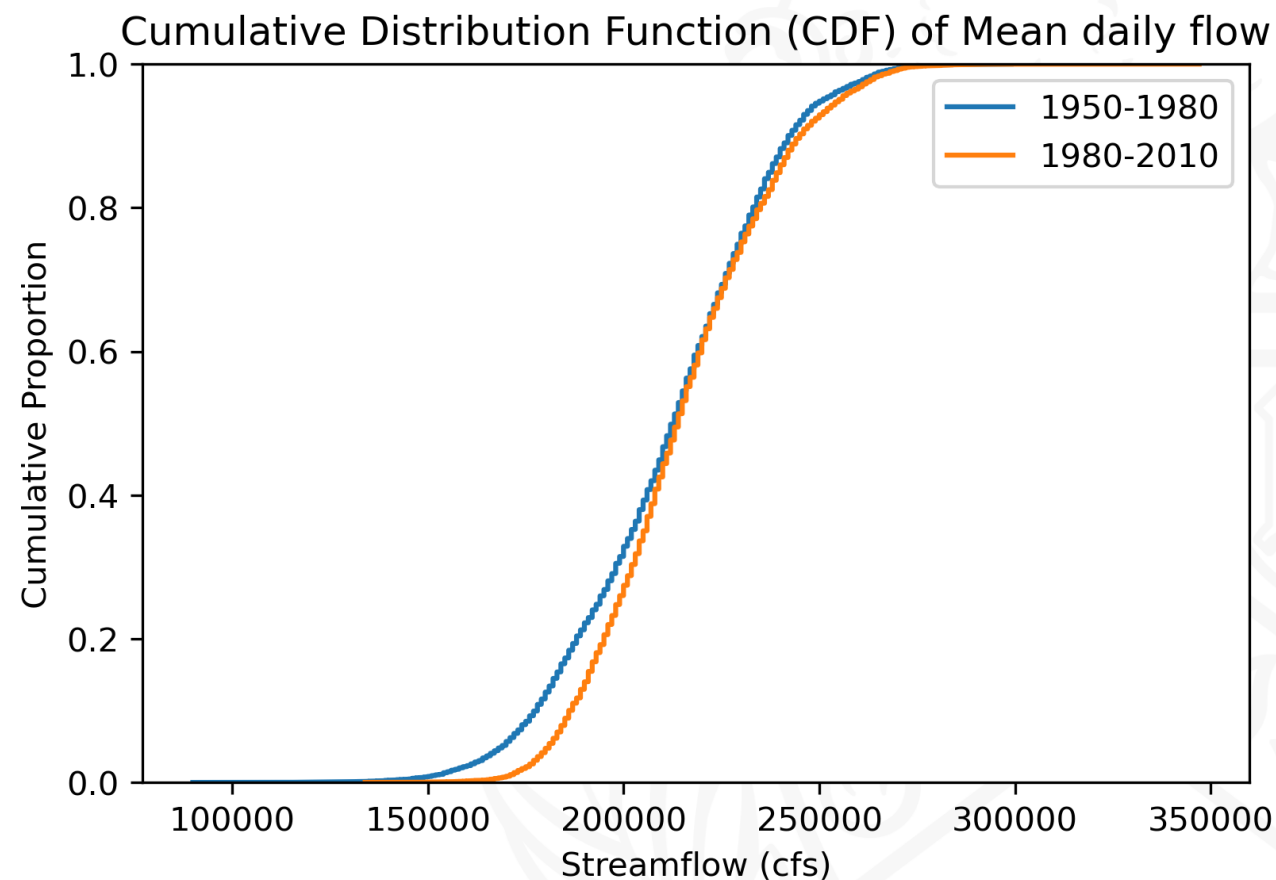
Use CDF to visualize the shift in hydrologic regimes



Use CDF to visualize the shift in hydrologic regimes



What information can we read from the plot in the right?



Questions

